Work involved in expansion and compression processes:

In most thermodynamic calculations we are dealing with the evaluation of work involved in the expansion or compression of gases. The essential condition for expansion or compression of a system is that there should be difference between external pressure (P_{ext}) and internal pressure (P_{int}).

For understanding pressure-volume work, let us consider a cylinder which contains 'n' moles of an ideal gas fitted with a frictionless piston of cross sectional area A. The total volume of the gas inside is V_i and pressure of the gas inside is P_{int}.

If the external pressure P_{ext} is greater than P_{int}, the piston moves inward till the pressure inside becomes equal to P_{ext}. Let this change be achieved in a single step and the final volume be V_f.

In this case, the work is done on the system (+w). It can be calculated as follows

$$w = -F \cdot \Delta x \quad \text{---------- (7.2)}$$

where dx is the distance moved by the piston during the compression and F is the force acting on the gas.

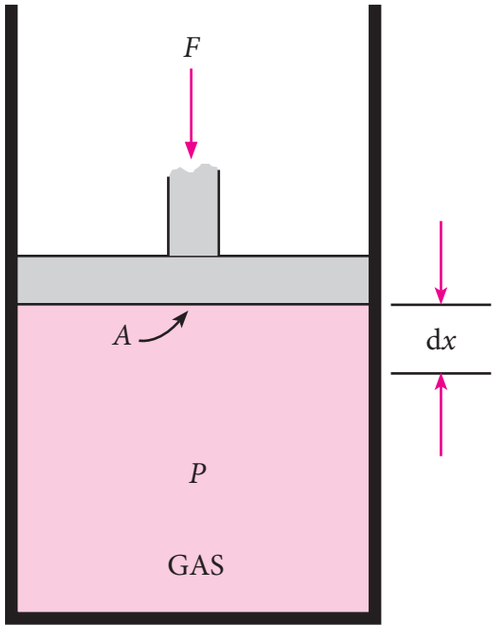

Figure 7.3 showing work involved in compression processes

$$F = P_{ext} A \text{---- (7.3)}$$

Substituting 7.3 in 7.2

$$w = -P_{ext} \cdot A \cdot \Delta x$$

$A.\Delta x$ = *change in volume* = $V_f - V_i$

$$w = -P_{ext} \cdot (V_f - V_i) \text{ ------ (7.4)}$$

$$w = -P_{ext} \cdot (-\Delta V) \text{ ------ (7.5)}$$

$$= P_{ext} \cdot \Delta V$$

Since work is done on the system, it is a positive quantity.

If the pressure is not constant, but changes during the process such that it is always infinitesimally greater than the pressure of the gas, then, at each stage of compression, the volume decreases by an infinitesimal amount, dV. In such a case we can calculate the work done on the gas by the relation

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